

# API Reference

The versioning rule, authentication, the shared response envelopes, DTO conventions, and the full endpoint reference.

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The Finance Portal exposes a versioned REST API behind the Nginx gateway. This document is the contract: the versioning rule, how authentication works, the response envelopes every endpoint shares, the DTO conventions, and the full endpoint map. Errors have their own document — see *Error Handling*.

## Versioning

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Every endpoint is served under a URI version prefix:

```
/api/v1/<resource>
```

- The prefix ( `v1` ) is the **major** API version. Breaking changes — a removed field, a changed type, a different status code — ship under a new prefix ( `/api/v2` ) so existing clients keep working. Additive, backward-compatible changes stay in `v1`.
- The product itself is released with semantic versioning ( `v0.x` during development); the API prefix tracks only the wire contract, independent of the product version.
- The gateway routes `/api/v1/{notifications,notification-preferences,price-alerts,watchlists,reports,market-status,...}` to the notification service ( `:8082` ) and everything else under `/api/v1` to the core backend ( `:8080` ). Clients see one consistent base path.

## Interactive documentation (OpenAPI / Swagger)

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The API is described by an OpenAPI 3.1 spec generated from the controllers (Springdoc):

- **Swagger UI** (interactive, try-it-out): `http://localhost/swagger-ui/index.html`
- **OpenAPI spec** (machine-readable): `http://localhost/v3/api-docs` — a snapshot is kept at `docs/api/openapi.json`.

Each endpoint's **full contract** — **path & query parameters, request-body fields, validation rules, response schema and status codes** — **lives there**, generated from the controllers so it never drifts from the code. The endpoint map at the end of this document is the orientation layer; Swagger UI / `openapi.json` is the authoritative parameter reference.

Recurring query parameters: list endpoints page with `page` (0-indexed) and `size`, and many accept a `sort` / type / search filter — the exact set per resource is in Swagger.

## Authentication

The API is stateless and secured with **Keycloak (OIDC)**. Every request (except the two public endpoints noted below) carries a bearer token:

```
Authorization: Bearer <access_token>
```

- The token is a Keycloak JWT; the backend validates it against the realm's JWKS.
- The caller is identified by the token's `sub` claim — all user-scoped data (portfolios, alerts, preferences) is keyed on it.
- Admin endpoints additionally require the realm role `ADMIN` ( `@PreAuthorize` ).
- Public endpoints: `GET /api/v1/markets/funds/sub-categories` and `GET /api/v1/notifications/ping`.

The gateway also rate-limits: ~50 req/s for general traffic, ~5 req/s on auth-sensitive routes.

## Response envelope — `ApiResponse<T>`

Every successful response is wrapped in one envelope, so clients parse a single shape. Controllers never return entities directly.

```
public class ApiResponse<T> {
    private boolean success;           // true
    private String message;           // optional, localized
    private T data;                   // payload: a DTO, a PagedResponse<T>, a collection, or a
    scalar
    private LocalDateTime timestamp;
}
```

```
{
  "success": true,
  "message": "Portfolio created",
  "data": { "id": 12, "name": "Ana Portföy", "createdAt": "2026-05-29T10:15:30" },
  "timestamp": "2026-05-29T10:15:30"
}
```

## Pagination — `PagedResponse<T>`

List endpoints wrap their items in a `PagedResponse<T>`, nested inside `ApiResponse.data`:

```
public record PagedResponse<T> (
    List<T> content, int page, int size, long totalElements, int totalPages) {}
```

```
{
  "success": true,
  "data": { "content": [ /* ... */ ], "page": 0, "size": 20, "totalElements": 137, "totalPages": 7 },
  "timestamp": "2026-05-29T10:15:30"
}
```

## Errors — ErrorResponse

Failures use a parallel `ErrorResponse` envelope ( `success: false`, `errorCode`, `message`, `path`, ...) so success and error paths are handled uniformly on the client. The full field list and the exception → HTTP status map are in the **Error Handling** document.

## DTO & mapping conventions

```
com.finance.{module}.dto
├─ request/      # input DTOs – suffix *Request, bean-validation annotated
└─ response/     # output DTOs – suffix *Response
```

- DTOs are immutable Java **records**; entities never cross the controller boundary.
- Request DTOs validate declaratively ( `@NotNull`, `@NotBlank`, `@Positive` ); a violation becomes a `400 VALIDATION_ERROR` with a per-field map.
- No `Map<String, Object>` on the response surface — type-specific data uses **sealed** records (e.g. `MarketAssetMetadata` permits `StockMetadata`, `FundMetadata`, ...).
- Entity → DTO mapping is done with **MapStruct** ( `@Mapper(componentModel = "spring")` ); computed fields (live price, P&L) are passed in by the service. `@AfterMapping` hooks enforce monetary scaling ( `scale = 4` money, `8` quantity ).

```
public record PositionRequest(
    @NotBlank String assetType, @NotBlank String assetCode,
    @NotNull @Positive BigDecimal quantity,
    @NotNull LocalDateTime entryDate, @NotNull @Positive BigDecimal entryPrice,
    LocalDateTime exitDate, @Positive BigDecimal exitPrice,
    String priceCurrency /* ISO-4217; null = already TRY */) {}
```

## Endpoint map

All paths are relative to `/api/v1`. Authenticated unless marked **Public**; **Admin** needs the `ADMIN` role.

### User & profile

| METHOD                                 | PATH                                                                                | PURPOSE                                 |
|----------------------------------------|-------------------------------------------------------------------------------------|-----------------------------------------|
| <code>GET</code> / <code>PUT</code>    | <code>/user/profile</code>                                                          | Get / update the current user's profile |
| <code>POST</code>                      | <code>/user/credentials/password/initiate-change</code>                             | Start password reset (Keycloak email)   |
| <code>POST</code>                      | <code>/user/credentials/email/initiate-change</code> · <code>/confirm-change</code> | Email change flow                       |
| <code>GET</code> / <code>DELETE</code> | <code>/user/credentials/2fa</code> · <code>/2fa/devices/{id}</code>                 | 2FA status / disable / remove device    |
| <code>GET</code> / <code>PATCH</code>  | <code>/user/preferences</code>                                                      | Read / partially update preferences     |

| METHOD    | PATH                                 | PURPOSE                          |
|-----------|--------------------------------------|----------------------------------|
| GET / PUT | /user/layout · /user/layout/overview | Dashboard layout                 |
| GET / PUT | /user/chart-data/{type}/{code}/...   | Per-asset chart prefs + drawings |

## Portfolios & positions

| METHOD       | PATH                                               | PURPOSE                                   |
|--------------|----------------------------------------------------|-------------------------------------------|
| GET / POST   | /portfolios                                        | List / create portfolios                  |
| PUT / DELETE | /portfolios/{id}                                   | Rename / delete                           |
| GET          | /portfolios/{id}/view                              | Combined summary + positions + allocation |
| GET          | /portfolios/{id}/summary · /allocation             | Totals & returns · allocation breakdown   |
| GET / POST   | /portfolios/{id}/positions                         | List (paged) / add a spot position        |
| PUT / DELETE | /portfolios/{id}/positions/{pid}                   | Update / delete a position                |
| POST         | /portfolios/{id}/positions/{pid}/sell · /reopen    | Close / reopen                            |
| GET          | /portfolios/{id}/backfill-stream                   | Snapshot backfill progress (SSE)          |
| GET / POST   | /portfolios/{id}/derivatives                       | VIOP derivative positions                 |
| PATCH / PUT  | /portfolios/{id}/derivatives/{pid}/close · /reopen | Close / amend / reopen                    |

## Market, instruments & analytics

| METHOD | PATH                                                      | PURPOSE                            |
|--------|-----------------------------------------------------------|------------------------------------|
| GET    | /market                                                   | Unified cross-asset search (paged) |
| GET    | /market/history · /group-counts · /availability           | History · counts · coverage        |
| GET    | /market/overview · /overview/widget-definitions           | Rendered widgets · catalog         |
| GET    | /market-status                                            | Open/closed session per market     |
| GET    | /bonds · /bonds/{seriesCode} · /bonds/rate-history/{isin} | Bonds + rate history               |
| GET    | /macro-indicators · /{code}/history                       | Macro indicators                   |
| GET    | /bank-rates · /bank-rates/currencies                      | Exchange / asset rates             |
| GET    | /tracked-assets · /{type}/{code}                          | Tracked-asset registry             |
| POST   | /analytics/scenarios                                      | Scenario simulation                |
| GET    | /analytics/inflation-beaters · /portfolio-series/{id}     | Inflation ranking · value series   |

## News & reports

| METHOD | PATH                                  | PURPOSE                         |
|--------|---------------------------------------|---------------------------------|
| GET    | /news · /news/categories · /news/{id} | List / categories / detail      |
| POST   | /reports/portfolio-pdf                | Generate a portfolio PDF report |

## Notifications, alerts & watchlists (*notification service, :8082*)

| METHOD                    | PATH                                           | PURPOSE                    |
|---------------------------|------------------------------------------------|----------------------------|
| GET                       | /notifications · /unread-count · /stream (SSE) | List / count / live stream |
| PATCH / POST / DELETE     | /notifications/{id}/read · /read-all · /{id}   | Read / bulk read / delete  |
| GET / PATCH               | /notification-preferences                      | Read / update preferences  |
| POST / GET / PUT / DELETE | /price-alerts ...                              | Manage price alerts        |
| GET / POST / PUT / DELETE | /watchlists · /watchlists/{id}/items ...       | Manage watchlists          |

## Admin

| METHOD                      | PATH                                          | PURPOSE                         |
|-----------------------------|-----------------------------------------------|---------------------------------|
| GET / PUT                   | /admin/users · /{id}/ban · /unban             | Keycloak user management        |
| GET / POST / PATCH / DELETE | /admin/tracked-assets · /admin/news-sources   | Reference data management       |
| POST                        | /admin/trigger/{type}/{snapshot candles full} | Trigger a data refresh (202)    |
| GET                         | /admin/tasks/stream                           | Live task status (SSE)          |
| POST                        | /admin/notifications/broadcast                | Broadcast a system notification |

The exhaustive, always-current list (request/response schemas, every field) lives in Swagger UI and `openapi.json` — this map is the orientation layer on top of it.

## Parameters reference

Generated from the OpenAPI spec. `*` = required. Path params appear in the path; **Body** names a request DTO (fields are in the DTO section / Swagger).

### admin-controller

| ENDPOINT                            | QUERY PARAMS | BODY |
|-------------------------------------|--------------|------|
| POST /admin/trigger/{type}/snapshot | —            | —    |
| POST /admin/trigger/{type}/full     | —            | —    |

| ENDPOINT                           | QUERY PARAMS | BODY |
|------------------------------------|--------------|------|
| POST /admin/trigger/{type}/candles | —            | —    |
| POST /admin/trigger/news/update    | —            | —    |
| POST /admin/trigger/macro/refresh  | —            | —    |
| POST /admin/trigger/bond/update    | —            | —    |
| GET /admin/tasks/stream            | —            | —    |

## admin-news-source-controller

| ENDPOINT                               | QUERY PARAMS              | BODY                    |
|----------------------------------------|---------------------------|-------------------------|
| GET /admin/news-sources/{id}           | —                         | —                       |
| PUT /admin/news-sources/{id}           | —                         | UpsertNewsSourceRequest |
| DELETE /admin/news-sources/{id}        | —                         | —                       |
| GET /admin/news-sources                | includeDisabled (boolean) | —                       |
| POST /admin/news-sources               | —                         | UpsertNewsSourceRequest |
| PATCH /admin/news-sources/{id}/enabled | enabled* (boolean)        | —                       |

## admin-tracked-asset-controller

| ENDPOINT                                   | QUERY PARAMS                                                      | BODY                               |
|--------------------------------------------|-------------------------------------------------------------------|------------------------------------|
| GET /admin/tracked-assets                  | type (string), search (string), sort (string), direction (string) | —                                  |
| POST /admin/tracked-assets                 | —                                                                 | UpsertTrackedAssetRequest          |
| PATCH /admin/tracked-assets/order          | —                                                                 | BulkTrackedAssetOrderUpdateRequest |
| DELETE /admin/tracked-assets/{type}/{code} | —                                                                 | —                                  |

## admin-user-controller

| ENDPOINT                    | QUERY PARAMS                                    | BODY |
|-----------------------------|-------------------------------------------------|------|
| PUT /admin/users/{id}/unban | —                                               | —    |
| PUT /admin/users/{id}/ban   | —                                               | —    |
| GET /admin/users            | first (integer), max (integer), search (string) | —    |
| GET /admin/users/count      | search (string)                                 | —    |

## analytics-controller

| ENDPOINT                                      | QUERY PARAMS                                                 | BODY            |
|-----------------------------------------------|--------------------------------------------------------------|-----------------|
| POST /analytics/scenarios                     | —                                                            | ScenarioRequest |
| GET /analytics/portfolio-series/{portfolioId} | from* (string), to* (string)                                 | —               |
| GET /analytics/inflation-beaters              | period (string), benchmark (string), targetCurrency (string) | —               |

## bank-rates-controller

| ENDPOINT                   | QUERY PARAMS                     | BODY |
|----------------------------|----------------------------------|------|
| GET /bank-rates            | currency (string), kind (string) | —    |
| GET /bank-rates/currencies | kind (string)                    | —    |

## bond-controller

| ENDPOINT                           | QUERY PARAMS                                                                                          | BODY |
|------------------------------------|-------------------------------------------------------------------------------------------------------|------|
| GET /bonds                         | search (string), bondType (string), sort (string), direction (string), page (integer), size (integer) | —    |
| GET /bonds/{seriesCode}            | —                                                                                                     | —    |
| GET /bonds/types                   | —                                                                                                     | —    |
| GET /bonds/rate-history/{isinCode} | period (string)                                                                                       | —    |

## derivative-position-controller

| ENDPOINT                                                        | QUERY PARAMS       | BODY                            |
|-----------------------------------------------------------------|--------------------|---------------------------------|
| PUT /portfolios/{portfolioId}/derivatives/{positionId}          | —                  | UpdateDerivativePositionRequest |
| DELETE /portfolios/{portfolioId}/derivatives/{positionId}       | —                  | —                               |
| PUT /portfolios/{portfolioId}/derivatives/{positionId}/close    | —                  | CloseDerivativePositionRequest  |
| PATCH /portfolios/{portfolioId}/derivatives/{positionId}/close  | —                  | CloseDerivativePositionRequest  |
| GET /portfolios/{portfolioId}/derivatives                       | openOnly (boolean) | —                               |
| POST /portfolios/{portfolioId}/derivatives                      | —                  | OpenDerivativePositionRequest   |
| PATCH /portfolios/{portfolioId}/derivatives/{positionId}/reopen | —                  | —                               |

## fund-detail-controller

| ENDPOINT                          | QUERY PARAMS | BODY |
|-----------------------------------|--------------|------|
| GET /markets/funds/sub-categories | —            | —    |

## macro-indicator-controller

| ENDPOINT                             | QUERY PARAMS                               | BODY |
|--------------------------------------|--------------------------------------------|------|
| GET /macro-indicators                | category (string), prominentOnly (boolean) | —    |
| GET /macro-indicators/{code}/history | from (string), to (string)                 | —    |

## market-overview-controller

| ENDPOINT                                | QUERY PARAMS  | BODY |
|-----------------------------------------|---------------|------|
| GET /market/overview                    | page (string) | —    |
| GET /market/overview/widget-definitions | —             | —    |

## news-controller

| ENDPOINT             | QUERY PARAMS                                                                                          | BODY |
|----------------------|-------------------------------------------------------------------------------------------------------|------|
| GET /news            | category (string), search (string), sort (string), direction (string), page (integer), size (integer) | —    |
| GET /news/{id}       | —                                                                                                     | —    |
| GET /news/categories | —                                                                                                     | —    |

## portfolio-controller

| ENDPOINT                                                | QUERY PARAMS | BODY                   |
|---------------------------------------------------------|--------------|------------------------|
| PUT /portfolios/{portfolioId}                           | —            | PortfolioCreateRequest |
| DELETE /portfolios/{portfolioId}                        | —            | —                      |
| PUT /portfolios/{portfolioId}/positions/{positionId}    | —            | PositionRequest        |
| DELETE /portfolios/{portfolioId}/positions/{positionId} | —            | —                      |
| GET /portfolios                                         | —            | —                      |
| POST /portfolios                                        | —            | PortfolioCreateRequest |

| ENDPOINT                                                                | QUERY<br>PARAMS                                                                                                                         | BODY                |
|-------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|---------------------|
| GET /portfolios/{portfolioId}/positions                                 | search<br>(string),<br>assetType<br>(string),<br>sort<br>(string),<br>direction<br>(string),<br>page<br>(integer),<br>size<br>(integer) | —                   |
| POST /portfolios/{portfolioId}/positions                                | —                                                                                                                                       | PositionRequest     |
| POST /portfolios/{portfolioId}/positions/{positionId}/sell              | —                                                                                                                                       | PositionSellRequest |
| POST /portfolios/{portfolioId}/positions/{positionId}/reopen            | —                                                                                                                                       | —                   |
| GET /portfolios/{portfolioId}/view                                      | include<br>(string)                                                                                                                     | —                   |
| GET /portfolios/{portfolioId}/summary                                   | assetType<br>(string)                                                                                                                   | —                   |
| GET /portfolios/{portfolioId}/chart                                     | type*<br>(string),<br>range<br>(string),<br>assetType<br>(string),<br>assetCode<br>(string)                                             | —                   |
| GET /portfolios/{portfolioId}/backfill-stream                           | —                                                                                                                                       | —                   |
| GET<br>/portfolios/{portfolioId}/assets/{assetType}/{assetCode}/summary | —                                                                                                                                       | —                   |
| GET /portfolios/{portfolioId}/allocation                                | mode<br>(string),<br>assetType<br>(string),<br>limit<br>(integer)                                                                       | —                   |
| GET /portfolios/limits                                                  | —                                                                                                                                       | —                   |

## tracked-asset-controller

| ENDPOINT            | QUERY PARAMS                                                      | BODY |
|---------------------|-------------------------------------------------------------------|------|
| GET /tracked-assets | type (string), search (string), sort (string), direction (string) | —    |

| ENDPOINT                          | QUERY PARAMS | BODY |
|-----------------------------------|--------------|------|
| GET /tracked-assets/{type}/{code} | —            | —    |

## unified-market-controller

| ENDPOINT                 | QUERY PARAMS                                                                                                                                                                                                  | BODY |
|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| GET /market              | type (string), code (string), segment (string), search (string), subType (string), sort (string), direction (string), filter (string), page (integer), size (integer), subCategory (array), riskValue (array) | —    |
| GET /market/history      | type* (string), code* (string), period (string)                                                                                                                                                               | —    |
| GET /market/group-counts | type* (string)                                                                                                                                                                                                | —    |
| GET /market/availability | type* (string), code* (string), yearMonth* (string)                                                                                                                                                           | —    |

## user-chart-data-controller

| ENDPOINT                                       | QUERY PARAMS   | BODY                             |
|------------------------------------------------|----------------|----------------------------------|
| PUT /user/chart-data/{type}/{code}/preferences | —              | UserChartPreferenceUpdateRequest |
| PUT /user/chart-data/{type}/{code}/drawings    | —              | UserChartDrawingUpdateRequest    |
| GET /user/chart-data/{type}/{code}             | range (string) | —                                |

## user-credential-controller

| ENDPOINT                                            | QUERY PARAMS | BODY                          |
|-----------------------------------------------------|--------------|-------------------------------|
| POST /user/credentials/password/initiate-change     | —            | PasswordChangeInitiateRequest |
| POST /user/credentials/email/initiate-change        | —            | EmailChangeInitiateRequest    |
| POST /user/credentials/email/confirm-change         | —            | EmailChangeConfirmRequest     |
| GET /user/credentials/email/pending                 | —            | —                             |
| DELETE /user/credentials/email/pending              | —            | —                             |
| GET /user/credentials/2fa                           | —            | —                             |
| DELETE /user/credentials/2fa                        | —            | —                             |
| GET /user/credentials/2fa/devices                   | —            | —                             |
| DELETE /user/credentials/2fa/devices/{credentialId} | —            | —                             |

## user-layout-controller

| ENDPOINT                  | QUERY PARAMS | BODY     |
|---------------------------|--------------|----------|
| PUT /user/layout/overview | —            | JsonNode |

| ENDPOINT         | QUERY PARAMS | BODY |
|------------------|--------------|------|
| GET /user/layout | —            | —    |

user-preference-controller

| ENDPOINT                | QUERY PARAMS | BODY                        |
|-------------------------|--------------|-----------------------------|
| GET /user/preferences   | —            | —                           |
| PATCH /user/preferences | —            | UserPreferenceUpdateRequest |

user-profile-controller

| ENDPOINT          | QUERY PARAMS | BODY                 |
|-------------------|--------------|----------------------|
| GET /user/profile | —            | —                    |
| PUT /user/profile | —            | ProfileUpdateRequest |